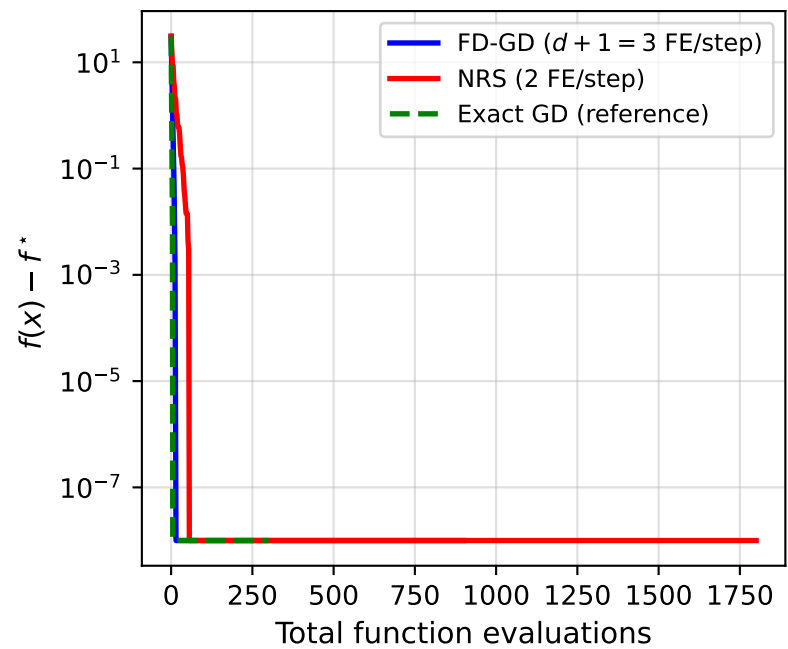
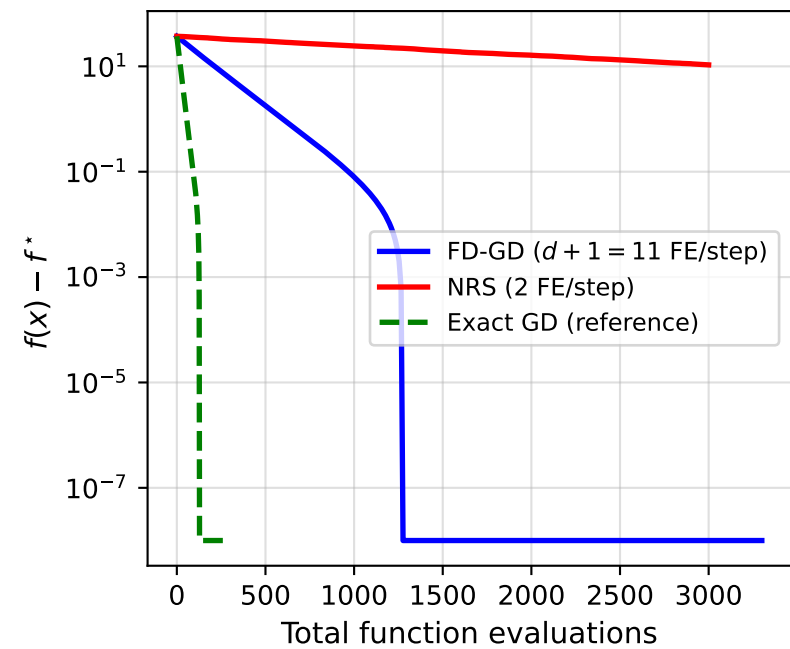


Q4: NRS vs FD-GD Oracle Complexity --- Dimension-Dependent Analysis

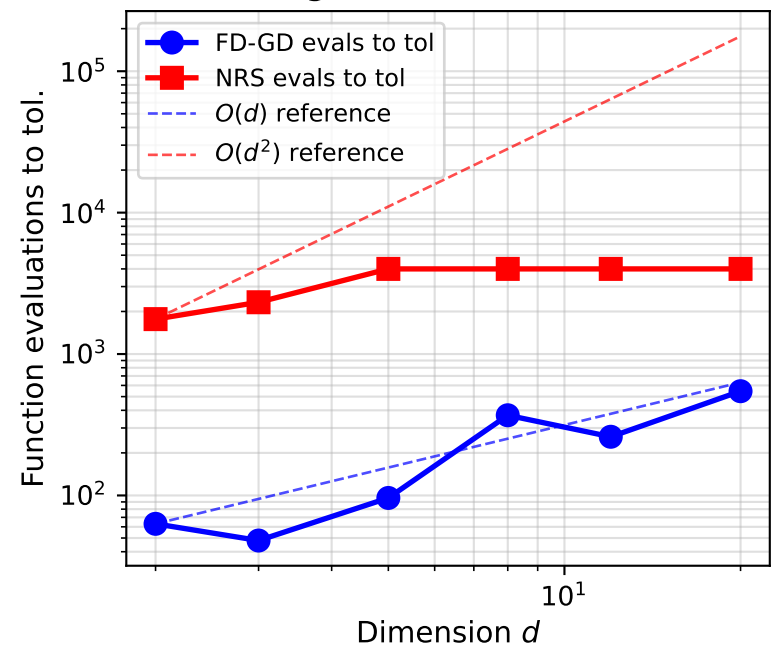
(a) $d = 2$: NRS vs FD-GD
($\kappa = 0.9$)



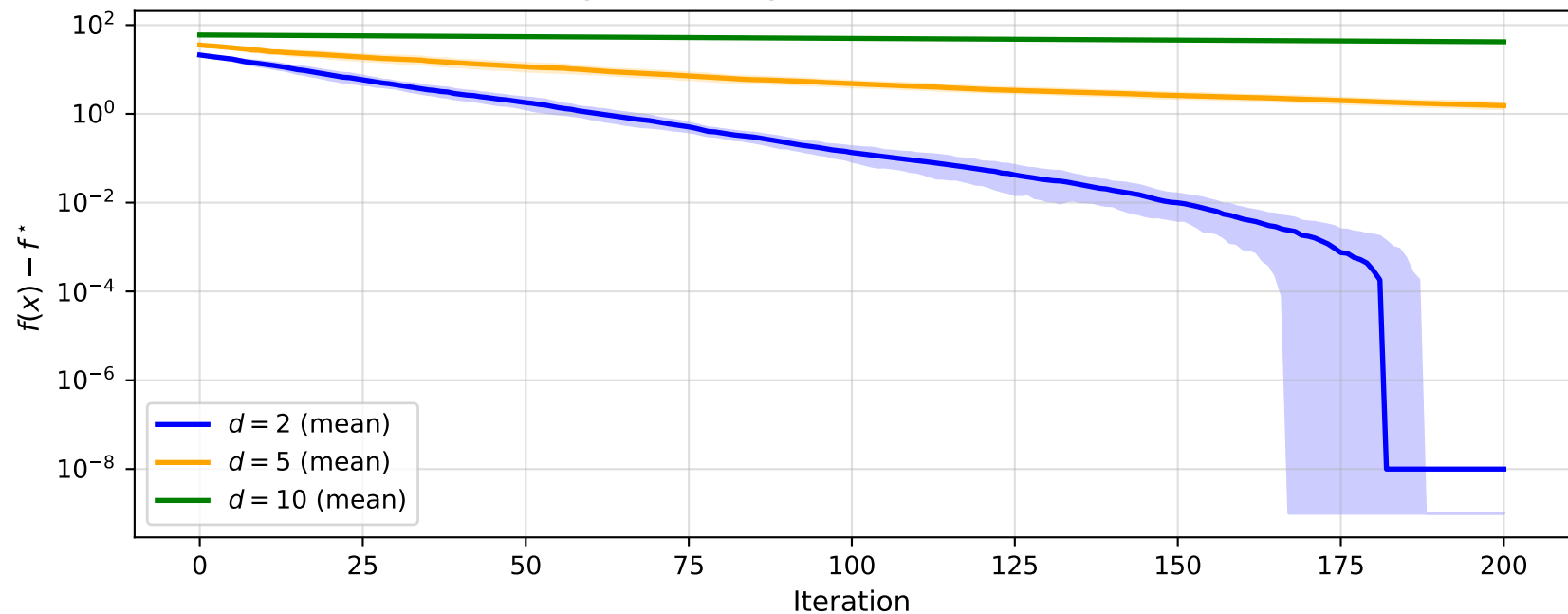
(b) $d = 10$: NRS disadvantage grows
($\kappa = 1.4$)



(c) Scaling: FD-GD $O(d)$ vs NRS $O(d^2)$



(d) NRS convergence vs dimension (mean ± 1 std over 8 runs)
Higher $d \rightarrow$ larger variance, slower rate



(e) Function evals to ε -accuracy
($\kappa = 10$, $\varepsilon = 10^{-3}$, factor of $\log(1/\varepsilon)$ omitted)

Method	$d = 2$	$d = 10$	$d = 100$	$d = 1000$
Exact GD	10	10	10	10
FD-GD	30	110	1010	10010
NRS	40	1000	100000	10000000